Abhay Sharma

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EDUCATION

University of Waterloo **Sep 2020 – Aug 2025**

- Bachelor of Mathematics (B.Math) Honours – Statistics Major, Computer Science & Economics Minor

SKILLS

Programming Languages: Python, R, SQL, Java, C#, TypeScript/JavaScript, C/C++, SAS

Tools & Frameworks: Jupyter, Docker, Kubernetes, OpenCV, Keras, Scitkit-learn, PyTorch, LangChain, Matplotlib, NumPy, Pandas, PySpark, PostgreSQL, Databricks, Power BI, Git, Linux, Azure, AWS, GCP

EXPERIENCE

Data Scientist **May 2024 – Dec 2024**

Innovation, Science and Economic Development Canada *Ottawa, ON*

- Optimized the ETL pipeline and designed LLM-based naming conventions for Beneficial Ownership Registry (BOR) using **Python** and **AWS (S3, Lambda, Redshift, Bedrock, Glue)**, boosting data retrieval accuracy by approximately **20%**.
- Implemented a custom AI Agent using **Python LangGraph** and **GPT-4** for NLP tasks in **AWS** (**EC2**, **SageMaker)**, leveraging the RAG framework to streamline departmental workflows, reducing manual workload by about **30%**.
- Deployed interactive dashboards using **Power BI** to visualize key performance metrics and trends, enabling data-driven decision-making within the department.

Data Engineer **May 2023 – Aug 2023**

Manulife Financial *Toronto, ON*

- Administered ServiceNow project migration to **Azure** by integrating 22/30 tables to an ETL pipeline utilizing **Azure Databricks** and **Blob Storage**, enhancing pipeline efficiency by **50% (from 4 hours to 2 hours)** using **PySpark, Jenkins,** and **JSON** triggers within **Azure Data Factory**, and resolving data load-related issues in the curation layer.
- Developed **Shell Scripts** for seamless migration of 450 table schemas across multiple environments from **Cloudera** to **Azure**, ensuring minimal downtime. Utilized **SQL** for data integrity checks, resolving migration-related incidents.

Data Science Engineer **Sep 2022 – Dec 2022**

Manulife Financial *Toronto, ON*

- Engineered a cost-saving data pipeline to integrate Canada's Deceased List (CDL) data into **Azure Data Lake Storage**, mapping it to annuities data using **Fuzzy Matching** logic, reducing company expenses by around **$400,000 annually**.
- Researched image detection models using **Azure's Labeling Tool**, **OpenCV**, and **Prodigy** for fraud detection, enhancing invoice verification processes by detecting fraudulent entries within **Azure Machine Learning Studio**.

Data Scientist **Jan 2022 – Apr 2022**

Natural Resources Canada | https://www.github.com/sharmabhay/modis250 *Ottawa, ON*

- Researched statistical methods and built ML models using **Random Forests** in **R** and **Neural Networks (ANN)** in **Python TensorFlow.Keras** for creating precise predictions with a model accuracy of **83%**.
- Created an algorithm capable of real-time mapping of global vegetation at 500m resolution to display MODIS data predictions geographically and enhance data visualization, utilizing **Google Earth Engine's JavaScript API**.

Software Developer I **May 2021 – Aug 2021**

Ukko Agro *Toronto, ON*

- Maintained and enhanced features in containerized **Flask REST API** middleware to expand the user database and compute more data points for users to access, increasing overall software efficiency by **15%**.

PROJECTS

Ruko | https://www.github.com/sharmabhay/ruko **Aug 2023 – Sep 2023**

- Created a Computer Vision-based color detection tool using **OpenCV**, **Pandas**, and **NumPy.**
- Plans to implement **K-Means Clustering** and **Naïve Bayes Classifier** for enhanced automation and accuracy in **Scikit-learn**, aiming to improve image recognition and classification tasks.